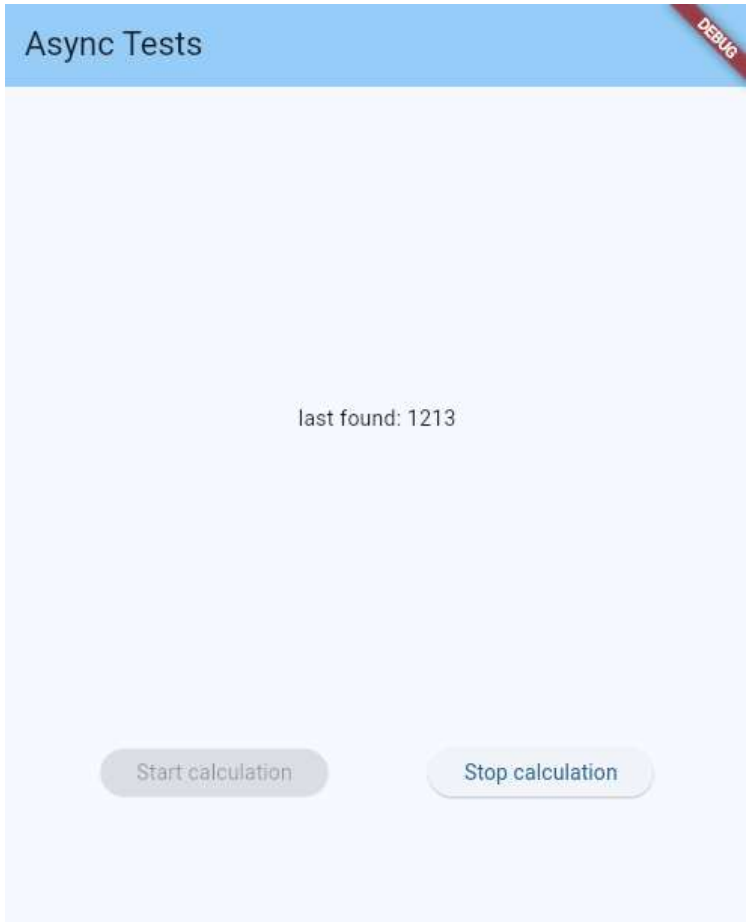
Interrupt long running callbacks in Flutter with asynchronous waits

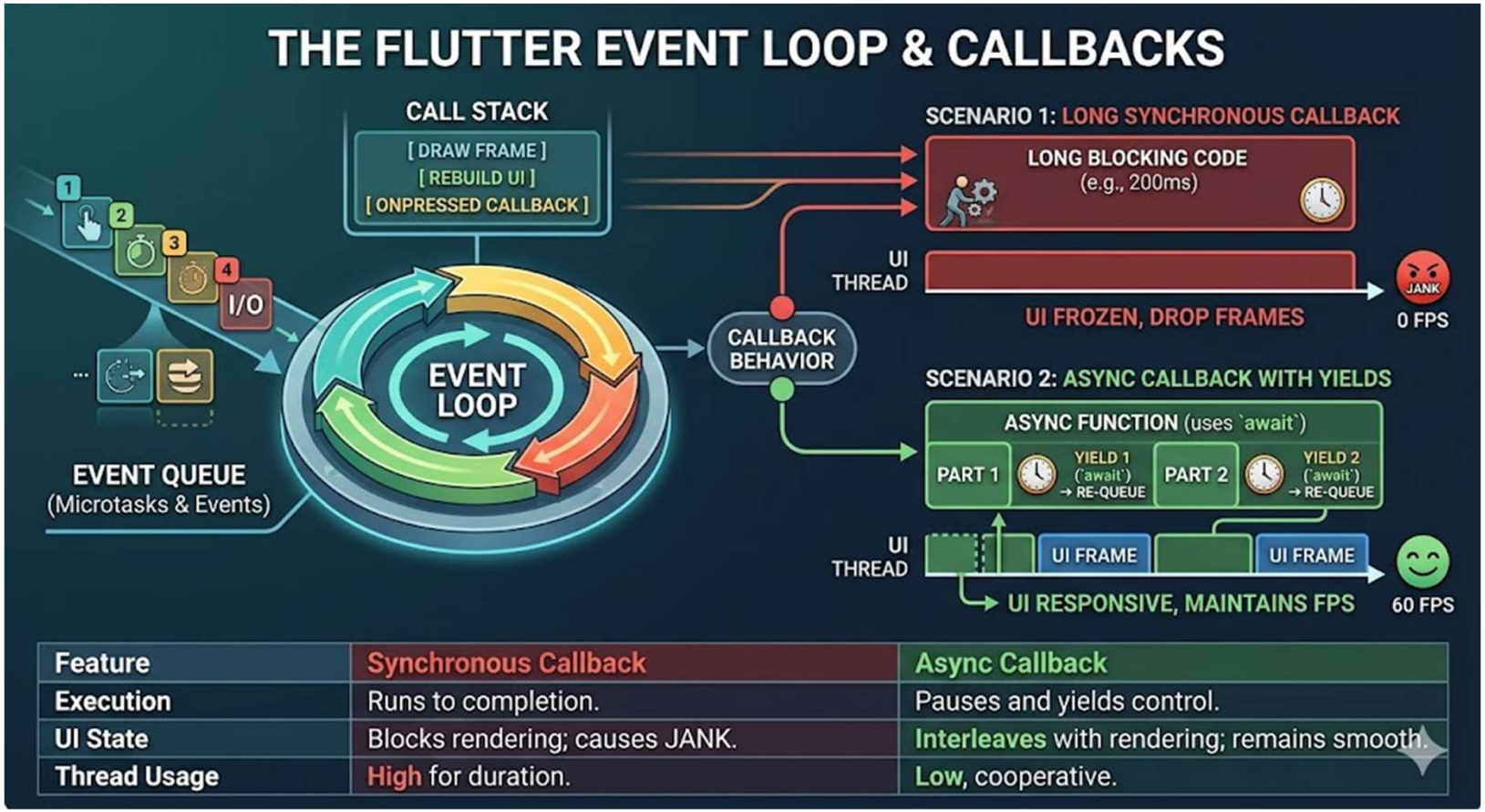
Find prime numbers between 5 and 2000




```
Future<int> performCalculation() async {
  var countFound = 0;
  for (int i = 5; i < 2000; i += 2) {
    if (isPrime(i)) {
      print("found prime $i");
      setState(() {
        lastFound = i;
      });
      countFound++;
      await Future.delayed(const Duration(milliseconds: 1));
      // next line would be a synchronous wait (not available in Chrome)
      //sleep(const Duration(milliseconds: 1));
    }
  }
  return countFound;
}
```

Web app under
https://guentherschmitt.github.io/web_test/async_test/

Event loop with async callbacks in Flutter



Feature	Synchronous Callback	Async Callback
Execution	Runs to completion.	Pauses and yields control.
UI State	Blocks rendering; causes JANK.	Interleaves with rendering; remains smooth.
Thread Usage	High for duration.	Low, cooperative.

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